

Temporally Constrained, Provenance-Verified Evidence Retrieval for Indian Legal Research

Yuvraj Singh, RishiRaj Jaiswal, Rahul Ranjan, Saurav Kumar Chaudhary, Kriti Mishra

Department of Computer Science and Engineering

Ajay Kumar Garg Engineering College

Ghaziabad, India

{yuvraj2312026, rishi2312003, rahul2312151, saurav2312023, kritimishra}@akgec.ac.in

Abstract—AI-assisted legal research can produce fluent answers with citations that do not exist, lack the attributed passage, post-date the matter, or were never retrieved. A 2026 Supreme Court of India ruling (2026 INSC 668) overturned tribunal findings built on exactly these defects. We present an evidence pipeline for historical Indian Supreme Court judgments in which every displayed citation is temporally eligible, provenance-bound, and mechanically verifiable, with outcome prediction kept separate. We report three main findings, supported by independent replay and source-side audit. First, naive ILDC-eCourts identifier matching proves unsafe: only 11 of 5,391 syntactic matches survived strict content alignment. Second, pre-ranking temporal eligibility raised Recall@5 from 5/30 to 12/30 and Recall@100 from 12/30 to 15/30 on the frozen 30-case base. Third, all 150 displayed base citations passed provenance and integrity verification with zero failed checks, while 15 expected authorities remained absent at $k=100$. Outcome baselines on 1,503 cases use their own population.

Index Terms—legal information retrieval, explainable AI, provenance verification, temporal filtering, Indian legal NLP

I. INTRODUCTION

Generated legal-research responses can cite nonexistent judgments, misattribute passages, or invoke unavailable law. Such hallucinations are frequent, not exceptional [1], and communicating legal standards to models remains open [2].

The problem is no longer confined to the literature. In *Pooja Ramesh Singh v. Jammu and Kashmir Bank Ltd. & Anr.*, 2026 INSC 668 (Civil Appeal No. 11950 of 2025, decided 2 July 2026), the Supreme Court of India set aside the underlying tribunal decisions after finding that several authorities relied upon in those proceedings could not be substantiated and that the cited material contained serious citation and content defects [3].

We treat that constraint as the design foundation: retrieved facts carry stable locators, duplicates are excluded, only verbatim passages are rendered, and each citation is verified against corpus and retrieval run.

All 150 displayed citations passed provenance and integrity verification, while 15 of the 30 expected authorities were absent from the top 100 results.

Contributions. We make four main contributions:

- 1) **A quantified cross-corpus identity failure and its remedy.** YYYY INSC N→YYYY_N conversion produced 5,391 syntactic candidates, of which only 11 passed content alignment (Section IV-C).
- 2) **Pre-ranking temporal eligibility.** Historical availability as a predicate inside the BM25 candidate relation, so

ineligible judgments cannot consume retrieval depth (Section VI-D): Recall@5 moved from 5/30 to 12/30 without regressing any prior success.

- 3) **A four-state citation protocol with separate denominators.** Fabricated, real-but-unretrieved, policy-violating, and valid-but-unexpected citations (Section VII).
- 4) **An evaluation that keeps recovery and integrity apart.** Authority recovery, citation integrity, and outcome prediction are reported on their own populations (Section VIII).

Scope. No legal correctness for retrieved authorities, no autonomous legal advisor, no claim that retrieval improves outcome prediction: a researcher-facing prototype. Two populations are evaluated and never pooled: 1,503 outcome cases and the 30 frozen evidence cases (Base-30).

II. RELATED WORK

A. Indian legal NLP and judgment prediction

ILDC established Indian Supreme Court judgment prediction and explanation as a benchmark task [4]; InLegalBERT [5] motivates our neural baseline; practitioner-evaluated summarisation [6] shows that legal documents are long and complex and that domain-expert evaluation can reveal aspects missed by document-wide metrics. None of these lines addresses cross-corpus identity or per-citation provenance, which is where this work sits.

B. Legal retrieval and temporal constraints

COLIEE evaluates legal case retrieval and entailment, while LegalBench provides a broad benchmark of legal reasoning tasks and CaseHOLD evaluates identification of relevant case holdings [7]–[9]. LegalBench-RAG further targets the retrieval step of legal RAG by evaluating precise source spans in legal corpora [10]. These benchmarks do not target the specific historical pre-dating eligibility constraint used here. TaxFlow adds validity filtering for Indian tax-law QA [11]; CaseFacts models temporal validity and changes in precedent [12], while LexTime evaluates temporal ordering of legal events [13]. Retrieval-augmented generation established retrieval as an explicit non-parametric evidence source [14]; BM25 supplies the sparse ranking basis [15] and Tesseract the OCR engine for corpus repair [16].

C. Positioning

We claim neither the first legal RAG system, nor the first Indian legal model, nor the first temporal legal benchmark: the contribution is the reproducible combination of content-alignment gating, conservative pre-ranking eligibility, exact retrieval-run verification, and separate recovery-versus-validity reporting on a source-first key. Jurisdictions, tasks, corpora, and denominators differ, so no external system is used as a numerical baseline. ILDC [4] supplies splits; InLegalBERT [5] motivates the neural baseline; TaxFlow [11] addresses statutory validity; CaseFacts [12] and LexTime [13] target validity and event ordering.

III. PROBLEM FORMULATION

Let a historical query judgment have identifier q , query year Y_q , and frozen facts-only text F_q . Let each candidate evidence passage p belong to a source judgment with a stable source identifier, an exact decision date, citation metadata, page and character boundaries, and stored passage text.

The temporal policy admits p only when the decision year of its source is strictly earlier than Y_q . A same-year source is treated as ambiguous, because ILDC supplies no exact query date; a source with missing or unparseable date metadata is excluded. Both rules trade recall so that, under the fail-closed metadata checks applied here, no displayed authority can post-date the matter.

The evidence task decomposes into three questions. Recovery asks whether the expected authority appears in the top 100 and among the five displayed. Integrity asks whether each citation is a real retrieved passage with exact provenance that avoids the query and duplicates and satisfies the temporal rule. Presentation asks whether the structure aids inspection; we do not evaluate it here.

Outcome prediction is secondary, reported on its own population and never pooled with evidence measures. Table I fixes definitions.

A. Research questions

We investigate two primary research questions. **RQ1:** Does pre-ranking temporal eligibility improve authority recovery (Recall@5, Recall@100) over the unfiltered BM25 baseline? Evidence-augmented outcome prediction on the 30-case set is reported descriptively alongside and is not directly comparable with the 1,503-case baselines (different population, task, and metric). **RQ2:** Does provenance-constrained selection and verification ensure that displayed evidence satisfies the defined grounding and integrity checks?

IV. CORPUS CONSTRUCTION AND AUDITING

A. Purpose-split corpus design

We use two corpora for different experimental purposes.

ILDC Single supplies fixed case-level splits and binary outcome labels: 7,593 judgments (5,082 training, 994 validation, 1,517 test). After a shared sufficiency rule excluded 14 test records, the prediction population contains 1,503 cases.

The evidence corpus is an eCourts-derived collection of Indian Supreme Court judgments covering 1950–2020. Of

TABLE I
OPERATIONAL DEFINITIONS (FROZEN EVALUATION).

Term	Implemented definition
Facts-only input	Text before the earlier of a recognised dispositive cue and a 60% character cap; inputs below 10% retention or 100 words are excluded
Temporal existence	The source has a parseable exact decision date sufficient to evaluate historical eligibility
Temporal applicability	For query year Y , an authority is eligible only if its decision year is $< Y$; same-year, later-year, and missing-date sources are excluded
Temporal enforcement	Eligibility is applied inside the BM25 candidate relation before ranking and <code>LIMIT 100</code>
Recall@100	Share of cases whose predefined authority occurs in the returned top-100
Recall@5	Share of cases whose predefined authority is among the five selected sources
Provenance validity	A displayed item reproduces the stored source ID, citation, decision date, court, locator, exact passage, and retrieval-run membership
Citation groundedness	Every displayed material proposition is verbatim text of a supplied passage linked to that evidence item
Authority consistency	A displayed source matches the answer-key authority by stable source ID, normalised citation, or title plus exact decision date

39,069 English PDF instances, 39,066 accepted PDFs yielded 2,343,435 labelled chunks, of which 2,036,981 unique chunks were loaded into a PostgreSQL provenance store and a SQLite FTS5 BM25 index.

This split follows the information actually available in each source. ILDC supports fixed-split outcome prediction but lacks citation and passage provenance; the eCourts collection supports dated, traceable retrieval but is not an outcome-label benchmark. Cross-corpus links are drawn only after the alignment checks described below.

B. Source-quality audit and targeted OCR repair

A full audit found valid UTF-8 text except 15 PDF instances (14 source IDs) with missing or corrupted embedded text.

Only flagged instances were re-rendered and processed with English Tesseract OCR at 250 DPI [16]. Instances passing a post-OCR quality gate were restored. Raw PDFs were preserved and the index rebuilt from the accepted corpus.

C. Cross-corpus identifier alignment

Linking ILDC query cases to eCourts sources requires establishing that two records describe the same judgment. The obvious route — converting an eCourts identifier of the form `YYYY INSC N` to the ILDC-like string `YYYY_N` — is unsafe, and quantifiably so.

That conversion produced 5,391 syntactic candidates. **Only 11 passed content alignment; 5,380 were identifier-namespace collisions.**

The corrected pipeline treats identifier equality as candidate generation only. A reusable gate evaluates title/party identity

with direct six-token phrase overlap before any mapping drives deduplication or exclusion. Combining syntactic and title/party candidates yielded 8,927 deduplicated pairs (1,304 accepted, 7,623 rejected). Retrieval additionally applies a full-document self-match safeguard, so an unmapped copy of the query judgment is removed without suppressing an earlier authority quoted at length by the later judgment.

D. Source-verified authority answer key

The authority evaluation uses 30 ILDC fixed-test cases. For each, the query judgment was inspected through a primary source or accepted eCourts mirror, one earlier authority was recorded *independently of system retrieval*, and reconciled by stable source ID, normalised citation, or title plus exact date.

The identifier-collision discovery triggered a read-only audit of the then-current mappings: 20 passed, nine failed direct content alignment, and one was unresolved. One case was relinked and the other nine replaced by new fixed-test cases, yielding 30/30 direct-content passes.

The sample is not era-balanced: five query cases are from the 1970s, 13 from the 1980s, nine from the 1990s, two from the 2000s, and one from the 2010s (1980s: 43.3%), a consequence of the source and alignment gates (Section X).

V. SYSTEM DESIGN

Four auditable layers implement the design. A corpus and alignment layer links ILDC cases to dated eCourts passages with content-checked deduplication. An outcome branch (E1 sparse linear, E2 InLegalBERT chunk-and-pool) predicts from facts only and never retrieves. An evidence branch (E3) retrieves BM25 passages under pre-ranking temporal eligibility and renders five source-diverse verbatim passages. A verification layer (E4) checks each displayed record against the corpus and the recorded retrieval run. The renderer is extractive and cannot retrieve, paraphrase material text, or infer outcomes; prediction stays separate.

VI. METHOD

A. Shared facts-only input

E1 and E2 share facts-only extraction (`ilcdc-predecision-facts-v1`): text preceding a dispositive cue and 60% of the document. Cases below 10% retention or 100 words are excluded; selection uses validation only, with one held-out test evaluation.

B. E1: sparse linear baseline

E1 uses lowercased TF-IDF unigrams/bigrams (sublinear TF, min DF two, L_2 , 100,000 features). Of $C \in \{0.1, 1.0, 10.0\}$, validation accuracy selected $C = 10.0$ (smaller C ties); refit once on eligible training plus validation and evaluated once on test. Seed 202605.

C. E2: InLegalBERT chunk-and-pool

E2 fine-tunes `law-ai/InLegalBERT` (revision `b5ecfed8`) with a new two-label head over overlapping 512-token windows with 50-token overlap. Training uses three epochs, seed 202607, learning rate 2×10^{-5} , weight decay 0.01, warm-up ratio 0.1, and gradient accumulation.

Mean pooling of window logits before softmax is the primary rule, and checkpoint selection uses validation document-level mean-logit accuracy only (majority vote is a secondary comparison, not a selection rule). All 1,503 eligible test documents and 9,576 test windows are covered.

D. E3: temporally constrained retrieval

The query builder (`tfidf-segment-salient-terms-v1`) emits at most 32 salient terms for a SQLite FTS5 BM25 index backed by PostgreSQL provenance.

The frozen configuration (`week11-bm25-salient-terms-preranked-temporal-v3`) admits only judgments with decision year strictly earlier than the query year into the candidate relation *before* BM25 ordering and the top-100 cutoff, so the full depth is spent on eligible material. The unfiltered BM25 baseline uses the same corpus, query builder, candidate generation, and top-100 cutoff as the temporal configuration, but applies no pre-ranking temporal eligibility predicate; all downstream alignment, self-match, and evidence-selection rules are held fixed.

Candidate sources are checked against the alignment-gated crosswalk and a self-match rule (100 shared six-token occurrences, 80% source-phrase coverage). A non-learned selector then chooses up to five passages in BM25 order, at most one per source.

E. E4: citation and provenance verification

E4 receives the E3 answer, retrieval-run identifier, and persisted corpus records. Each displayed item must exist in the corpus and reproduce its stored passage and provenance. It must also belong to the recorded run, exclude the query and duplicates, and pre-date the query year; any failure rejects the citation. Authority consistency is then evaluated separately: a fully valid citation may differ from the single expected authority without being substantively irrelevant (Section VIII-D).

F. Inference-time evidence augmentation

Outcome prediction was subsequently added to E3 and E4 by reusing the frozen E2 checkpoint without fine-tuning: the frozen facts extract plus the selected verbatim evidence in selection order pass through the same 512-token/50-overlap windows with mean-logit `argmax` (seed 202607; configuration `e3e4-evidence-augmented-inlegalbert-checkpoint6318-v1`). Gold labels, scores, dates, and provenance are excluded from model input.

Because the checkpoint was trained on facts-only inputs, applying it to facts plus retrieved passages is an inference-time distribution shift rather than evidence-aware fine-tuning. These predictions are reported descriptively on the frozen 30-case base, and are not directly comparable with the 1,503-case baselines.

G. Bundled-intervention caveat

E4 jointly checks corpus existence, passage/provenance identity, run membership, duplicate status, and temporal eligibility. Aggregate reliability cannot be attributed to any one component; the supported claim concerns the complete bundle.

VII. CITATION AND PROVENANCE PROTOCOL

Each chunk carries a stable identifier from its source path and passage location, with stored source ID, case ID, citation, exact decision date, court, PDF file, page and character bounds, and chunk text. Retrieval runs record query ID, year, query text, index version, temporal policy, rank, score, and temporal status.

The renderer may expose only supplied selected evidence. Material content remains verbatim; authority cards and conclusions refer to evidence IDs. Verification then checks corpus existence, exact field equality, run membership, duplicate status, and temporal eligibility, with answer-key matching only after a displayed citation has passed. Parallel reporter forms are reconciled through stable source identity or normalised title plus exact decision date, never loose string similarity.

This protocol distinguishes four failure states that a single “correct citation” label would collapse:

- 1) a fabricated or altered citation;
- 2) a real citation that was not retrieved for this query;
- 3) a retrieved citation violating temporal or duplicate policy;
- 4) a fully valid citation that does not match the predefined expected authority.

State (2) applies to a citation presented in output without corresponding retrieval-run membership; an expected authority absent from retrieval is instead counted as an authority-recovery miss.

In our results, states (1)–(3) occur zero times and state (4)—a fully valid citation not matching the single predefined expected authority—occurs in 18 of 30 *cases* (the 18 cases in which no selected citation matched the expected authority; the remaining 12 cases each contain the expected authority among the five selected).

VIII. RESULTS

A. Reporting populations

Two populations are reported and never pooled: 1,503 ILDC test cases (outcome prediction) and Base-30 (30 queries; 150 citations) for evidence evaluation.

B. Outcome prediction

On the 1,503-case population (Table II), sparse E1 reached accuracy 0.61344 and macro-F1 0.612342, exceeding E2 InLegalBERT mean-logit pooling at 0.596806 and 0.592358 (majority vote 0.6015 and 0.5937). On the 1,503-case population, E1 and E2 exceeded the majority-class accuracy of 0.5017. E3 and E4 are reported separately on the 30-case evidence population and are not compared with the 1,503-case majority baseline.

While this does not imply that domain pre-training is generally ineffective, it does show that the sparse baseline performed better under our frozen settings.

On the separate 30-case subset, E3 and E4 both reached accuracy 0.666667 and macro-F1 0.603175, with identical confusion matrices.

The equality is expected: with no evidence rejected, the predictor received an identical input.

TABLE II

OUTCOME PREDICTION (DISTINCT POPULATIONS; NOT A LEADERBOARD).

Model	n	Acc.	Macro-F1
Majority class	1,503	0.5017	0.3341
E1 TF-IDF + LR	1,503	0.61344	0.612342
E2 InLegalBERT, mean logits	1,503	0.596806	0.592358
E2 InLegalBERT, maj. vote	1,503	0.6015	0.5937
E3 evidence-augmented	30	0.666667	0.603175
E4 verified evid.-augmented	30	0.666667	0.603175

TABLE III

RQ1 AUTHORITY RECOVERY (BASE-30 REPLAYED).

Retrieval condition	n	R@5	R@100
Unfiltered BM25	30	0.1667 (5/30)	0.4000 (12/30)
Temporal pre-ranking	30	0.4000 (12/30)	0.5000 (15/30)

C. Authority recovery and evidence integrity

Base-30 recovery is Recall@5 0.40 (12/30) and Recall@100 0.50 (15/30: 12 selected + 3 retrieved-but-unselected); 15 authorities were absent at $k=100$.

The principal RQ1 result is the temporal pre-ranking row of Table III.

Table IV summarizes Base-30 integrity: 150/150 grounded and provenance-valid with 0/150 temporal violations and 0/30 cases with ungrounded displayed material.

E4 verification is fail-closed: any failing check rejects the citation. In the evaluated run, nothing was rejected, a ceiling effect of already-valid evidence.

The system verified all 150 displayed citations and recovered the expected authority in 15 of 30 cases at $k=100$.

D. Interpreting authority-consistency precision

Authority-consistency precision was 0.08, with recall 0.40 and F1 0.133: with one credited authority per case and five displayed, perfect inclusion caps the measure at $30/150 = 0.20$, so 0.08 is 40% of ceiling.

The 138 non-matching displayed citations are *not* independently annotated for substantive legal relevance. They passed every integrity check; they simply are not the one authority the key records.

E. Retrieval mechanism

The final configuration came from three recorded development steps on a nine-case probe (0/9 to 3/9 to 6/9 to 7/9 at $k=100$); pre-ranking temporal filtering, carried to the held-out base, moved Recall@5 5/30 to 12/30 and Recall@100 12/30 to 15/30 *without losing any of the 12 prior top-100 successes*. Probe history is reported from development records, not replayed.

Temporal capacity is not the whole story. 2013_35 stayed absent despite 79 eligible candidates in its raw top 100, while 1980_105 was recovered from only three and became a top-five hit: residual lexical mismatch is the prominent unresolved cause.

TABLE IV
EVIDENCE RECOVERY AND INTEGRITY, BASE-30 (30 QUERIES; 150 CITATIONS).

Measure	Value	Arithmetic
Recall@5	0.40	12 of 30 in top 5
Recall@100	0.50	15 of 30 in top 100
Absent at $k=100$	0.50	15 of 30
Authority precision	0.08	12 of 150 shown
struct. ceiling	0.20	30 of 150 (5 shown, 1 credited)
Grounding	1.00	150/150 passed
Provenance validity	1.00	150/150 passed
Temporal violations	0	0 of 150
Ungrounded displays	0	0 of 30 cases

IX. ERROR ANALYSIS

A. Outcome disagreement

The analysis is read-only over final frozen outputs. Across 1,503 cases, both models were correct on 684 and both wrong on 368; E1 alone was correct on 238 and E2 alone on 213 (on the 30-case key: 18 both correct, 3 E1-only correct, 3 E2-only correct, and 6 both wrong).

In the evidence-augmented run, E2 was wrong while E3 and E4 were correct for two cases (1974_36, 1984_136); E3/E4 verification changed neither the selected evidence nor the shared predictor input (0/30 divergences).

Critically, three cases (2008_1629, 1981_187, 1982_29) retrieved *and* selected the expected authority yet still produced an incorrect outcome prediction, and a fourth (1985_40) was retrieved but not selected yet likewise predicted incorrectly: correct recovery does not mechanically imply a correct prediction. No correct prediction was accompanied by a citation that failed the defined integrity checks.

B. Authority-recovery buckets

The exhaustive partition is: 12 retrieved and selected; 3 retrieved but unselected (1980_133 rank 15, 1981_55 rank 28, 1985_40 rank 78); and 15 absent at $k=100$. The middle bucket isolates *selection* failure (available but undisplayed), while the absent bucket requires upstream improvement.

X. LIMITATIONS AND THREATS TO VALIDITY

Scope. English-language Indian Supreme Court judgments only.

Human-rated explanation quality. Human-rated explanation quality was not evaluated in this study. Explainability is therefore assessed operationally through evidence linkage, provenance, and citation verification rather than through a human-rating study.

Extraction approximation. Some affected PDF instances were restored through the post-OCR quality gate [16]; repaired text may retain noise. With no gold facts/reasoning boundary, the facts rule is approximate.

Answer-key size and era concentration. Evidence evaluation rests on 30 frozen replayed cases; 13 of the base 30

are from the 1980s. The evaluated set is neither large nor era-balanced and should not be read as representative of the full ILDC test set or of Indian legal research generally.

Single reference authority. One expected authority per query, not every relevant source: inconsistency is a reference mismatch, not a legal-relevance judgment.

Residual recovery gap. At $k=100$, 15 of the 30 expected base authorities remain unrecovered; the 2013_35/1980_105 contrast shows temporal room alone does not surface the authority.

Year-level granularity. Query dates have year-level precision only, so same-year candidates are conservatively excluded.

Bundled verification. E4 checks integrity properties together and cannot isolate single-component contributions.

Inference numerics. Recomputed E3/E4 mean logits show bounded GPU fp16 variation across runs (bounded at the fourth decimal in the independent recomputation) that changed no decision, evidence selection, or metric; strict serialized file hashes of E3/E4 outputs therefore differ while substantive outputs agree.

Retrieval-equivalence scope. BM25 rebuild equivalence is established at ranking level — identical corpus, parameters, and top-100 agreement on all 30 evaluated queries — not as byte-identical SQLite files, and it does not generalize beyond the evaluated query set.

Prototype scale. Results apply to the frozen implementation and samples. No production deployment is claimed or evaluated.

XI. RESPONSIBLE USE AND GOVERNANCE

The system is a legal-research aid, not an adjudicator or provider of legal advice. E3/E4 predictions are experimental; a human reviewer judges relevance, currency, authority, and applicability [3]. Controls are fail-closed and negative results (discarded runs, collision correction, key replacements) are preserved.

XII. CONCLUSION AND FUTURE WORK

We designed, implemented, and evaluated a provenance-preserving, temporally constrained evidence workflow for historical Indian Supreme Court research.

The principal finding is a dissociation. Every one of 150 displayed base citations passed provenance and integrity verification with zero failed checks, while 15 of the 30 expected authorities were absent at $k=100$.

In this cross-corpus audit, only 11 of 5,391 syntactic identifier matches between ILDC and the eCourts collection were content-aligned.

We report supporting results along with their limits. The sparse TF-IDF baseline outperformed E2 InLegalBERT under the frozen settings (1,503 cases). Inference-time evidence augmentation reached 0.666667 accuracy and 0.603175 macro-F1 on the frozen 30-case subset, descriptively and under acknowledged distribution shift.

Future work. An era-balanced key of 75–100 cases for paired testing; blinded review with independent raters; hybrid dense–sparse retrieval for residual lexical misses; and evidence-sensitive uncertainty language.

REPRODUCIBILITY

All configurations, seeds, model revisions, corpus identities, and evaluation artifacts are frozen and machine-readable, including checkpoint-6318 (924a5bb9...dbdc773). The implementation passes 81 tests and the documented checks (14/14 documentation; freeze audit of 30 entries: 22 byte-exact, 8 byte-drift). All independently replayed primary metrics matched their frozen references; E3/E4 decisions, evidence selections, and metrics matched despite bounded fp16 variation, and BM25 matched at ranking level on the 30 evaluated queries. The demo stack is covered by 8/8 Spring Boot service tests and 6/6 FastAPI wrapper tests. Code and artifacts are available at <https://github.com/0-YuvrajSingh/NyayaTrace>.

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